

Certainty-Severity-Celerity: What Shapes Individual Protective Behaviour in Bacterial Epidemics?

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The Bacterial Burden

- ▶ India accounts for **25% of global tuberculosis cases** (WHO)
- ▶ **71.5% of healthy community members** carry drug-resistant Enterobacterales in their gut
- ▶ Bacterial keratitis affects **23.5% of corneal infection cases** in North India
- ▶ Pneumonia causes **47.1% in-hospital mortality** among infected elderly patients
- ▶ ICU mortality from resistant infections reaches **32%** (IMPRES study, 2025)

Antibiotic Consumption

- ▶ India is the **world's largest consumer of antibiotics** (ICMR data)
- ▶ At primary healthcare facilities in India, antibiotic consumption increased 49% annually between 2021-2023 accelerating selection pressure for resistance (TEPHINET study (2024))
- ▶ Over-the-counter availability without prescription remains common
- ▶ Irrational use in animal husbandry further amplifies resistance reservoirs

Antibiotics which are the primary defence are losing effectiveness at an alarming rate.

Vaccine Consumption

- ▶ 11 vaccines provided free under Universal Immunisation Programme one of world's largest public vaccine systems
- ▶ 98.6% full immunisation coverage achieved (2025) yet 1.6 million zero-dose children remain unvaccinated
- ▶ India lacks a national adult immunization schedule for ages 20–50. Unlike childhood programs, there is no routine government policy for bacterial vaccines (Pneumococcal, Typhoid, Tetanus) for healthy adults, leaving this population largely unprotected
- ▶ In Kerala (India's most literate state), 15% of the general population remained hesitant during COVID-19 despite high overall acceptance

Unlike antibiotics, vaccines face lower consumption due to access, awareness, and behavioral barriers..

Our Research Question

There is extensive biomedical data on resistance and consumption, but little understanding of what drives individual protective behaviour.

Research Question:

“What factors play a role in influencing people’s decision to adopt prophylactic measures i.e vaccination to prevent disease and curative antibiotics upon infection?”

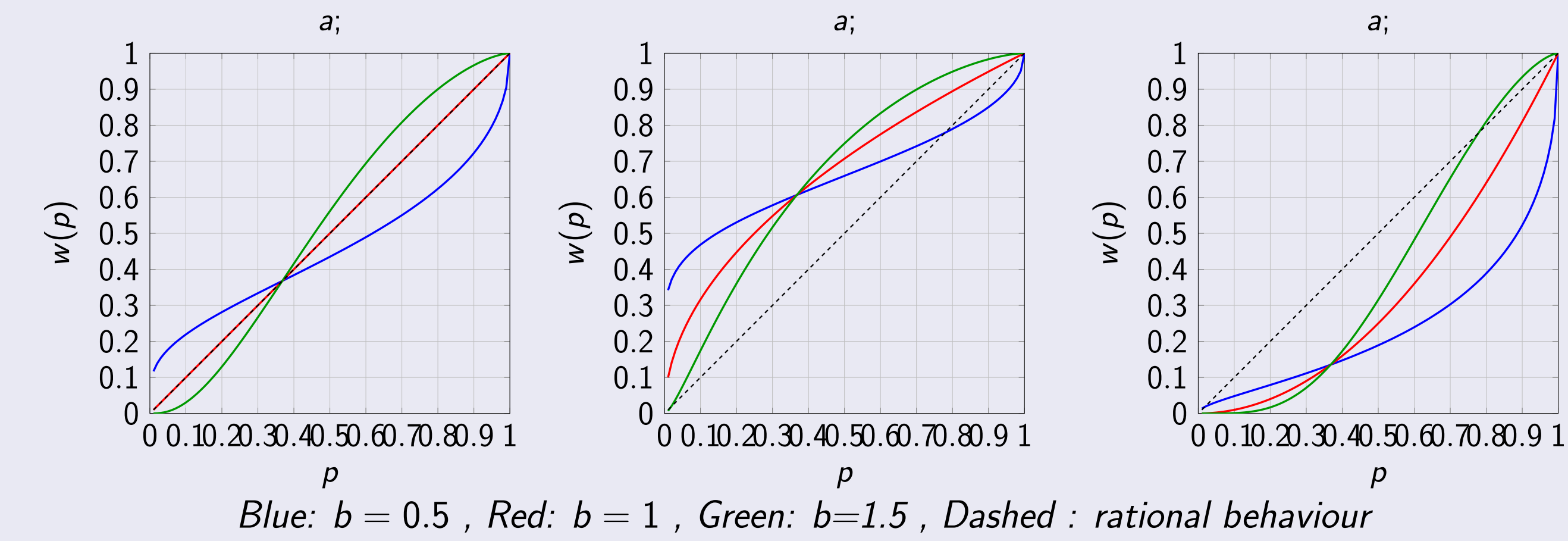
Theoretical lens: Certainty (risk perception) → Severity (outcome threat) → Celerity (speed of effective response)

Behavioral Biases

Mathematical Modeling of the Three Constructs:

- ▶ Certainty (Probability Weighting):
 - ▶ Prelec: $w(C) = \exp[-a(-\ln C)^b]$, $a, b > 0$
 - ▶ Applies to: π_S, π_D, ρ_R
 - ▶ Effect: Overweights small probabilities, underweights large ones
- ▶ Severity: $\pi_S^A = (1 - \epsilon_A)\pi_S$ (antibiotics reduce severity)
- ▶ Celerity (Temporal Discounting):
 - ▶ Hyperbolic: $\delta(\tau) = 1/(1 + k\tau)$
 - ▶ Delay ordering: $\tau_V \ll \tau_A \ll \tau_S \ll \tau_D \ll \tau_R$
 - ▶ Effect: Immediate costs weighted more than delayed consequences

Prelec Function: Transforms objective probability → subjective probability

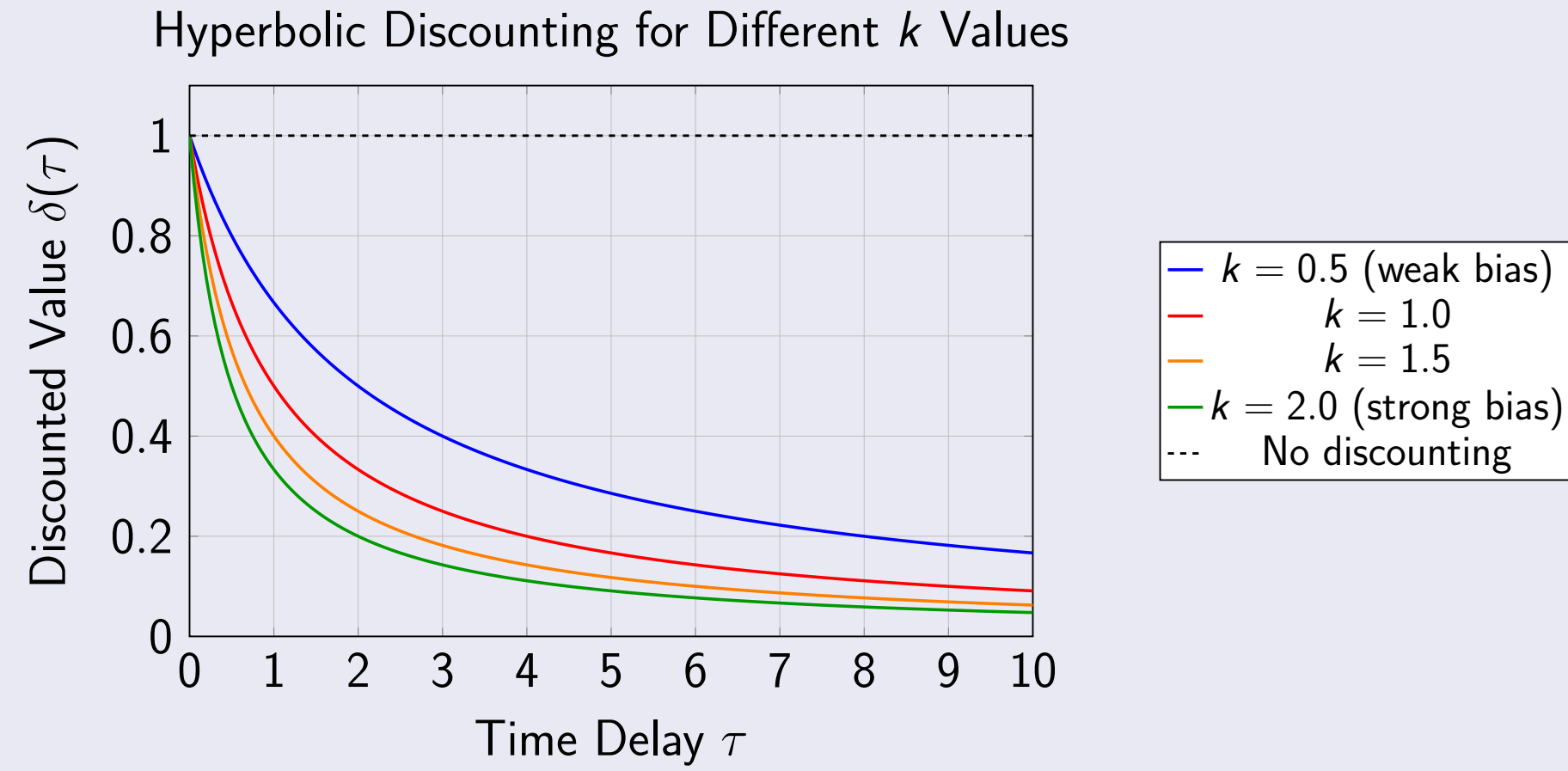


Temporal Discounting (Celerity)

Hyperbolic Discounting Function:

$$\delta(\tau) = \frac{1}{1 + k\tau}, \quad k > 0$$

Quantifies how individuals discount future consequences — higher k = stronger present bias.



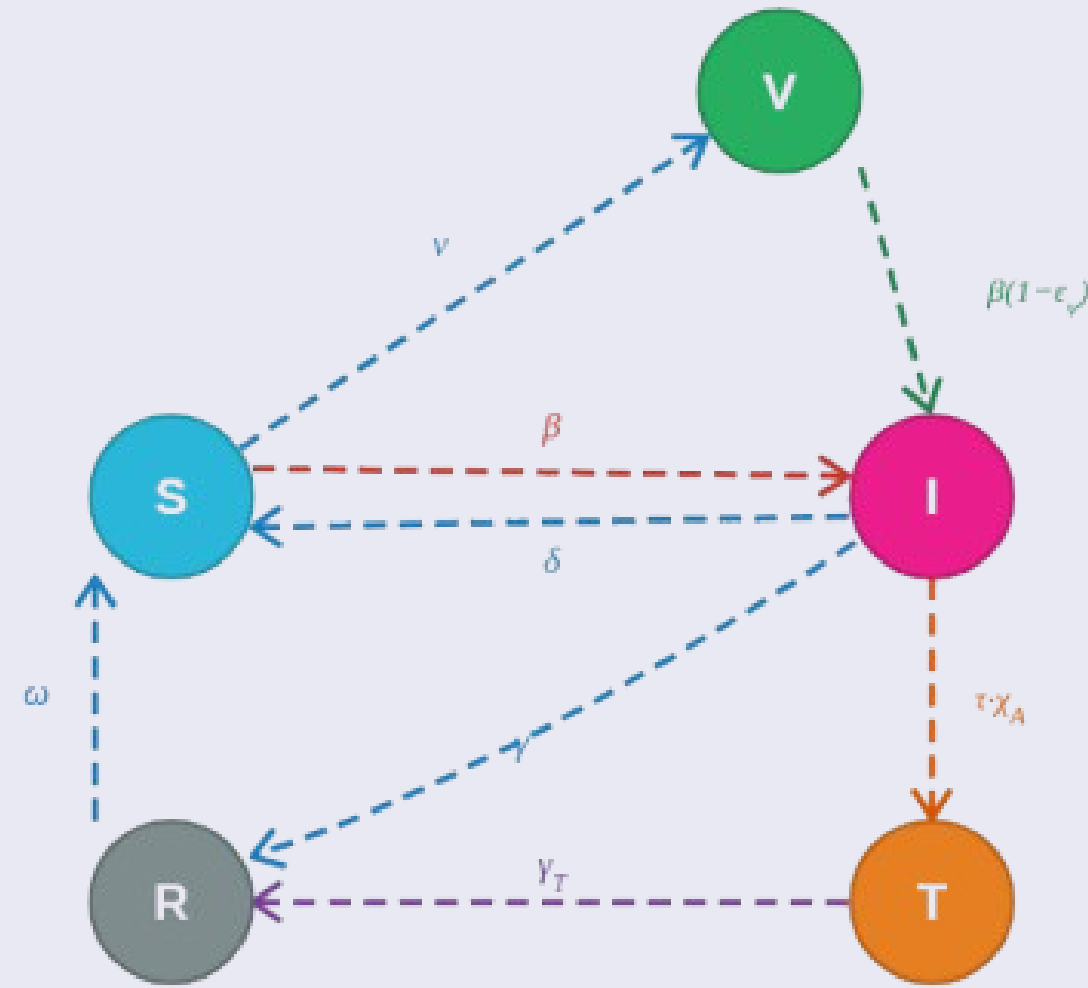
Key insight: Immediate costs (e.g., vaccine side effects) are weighted more heavily than delayed benefits (e.g., infection prevention), explaining under-vaccination.

SVITR Epidemiological Model

State Variables:

$$S + V + I + T + R = 1$$

- ▶ S : Susceptible (unvaccinated)
- ▶ V : Vaccinated susceptibles
- ▶ I : Infected (untreated)
- ▶ T : Infected (treated with antibiotics)
- ▶ R : Recovered



System of ODEs:

$$\begin{aligned} \dot{S} &= -\beta S(I + T) + \omega R - \nu x_V S \\ \dot{V} &= -(1 - \epsilon_V)\beta V(I + T) + \nu x_V S \\ \dot{I} &= \beta[S + (1 - \epsilon_V)V](I + T) - (\gamma + \tau x_A)I \\ \dot{T} &= \tau x_A I - \gamma_T T \\ \dot{R} &= \gamma I + \gamma_T T - \omega R \end{aligned}$$

Parameters: β (transmission), ϵ_V (vaccine efficacy), ν (vaccination rate), τ (treatment rate), γ (recovery without treatment), γ_T (recovery with treatment), ω (waning immunity)

Strategic Decision-Making: Payoffs & Evolution

Three competing strategies:

- ▶ V : Vaccination (prophylactic)
- ▶ A : Antibiotic reliance (curative upon infection)
- ▶ N : No protection

Payoff Functions :

- ▶ $f_V = -[\delta(\tau_V)c_V + \lambda_V\delta(\tau_S)(w(\pi_S)L_S + w(\pi_S)w(\pi_D)L_D)]$
- ▶ $f_A = -[\delta(\tau_A)c_A + \lambda w(\pi_S^A)\delta(\tau_S)L_S + \lambda w(\pi_S^A)w(\pi_D)\delta(\tau_D)L_D + w(\rho_R)\delta(\tau_R)\Omega]$
- ▶ $f_N = -\lambda\delta(\tau_S)(w(\pi_S)L_S + w(\pi_S)w(\pi_D)L_D) - \lambda c_N$

λ = force of infection, λ_V = force of infection(vaccinated individuals)

Replicator Dynamics :

$$\dot{x}_i = 0.1 x_i (f_i - \bar{f}), \quad i \in \{V, A, N\}$$

where $\bar{f} = x_V f_V + x_A f_A + x_N f_N$ is the average population payoff.

Eco-evolutionary coupling:

- ▶ Epidemiological states (S, V, I, T, R) influence perceived payoffs
- ▶ Strategy frequencies (x_V, x_A, x_N) modify disease transmission and recovery

Basic Reproduction Number R_0

Definition: Average number of secondary infections caused by a single infected individual in a completely susceptible population.

Derived expression :

$$R_0 = \frac{\beta[S + (1 - \epsilon_V)V](\gamma_T + \tau x_A^*)}{\gamma_T(\gamma + \tau x_A^*)}$$

At Disease-Free Equilibrium (DFE): $x_A^* = 0$

$$R_0 = \frac{\beta[S + (1 - \epsilon_V)V]}{\gamma}$$

Two limiting cases:

- ▶ No vaccination ($x_V = 0, S = 1, V = 0$):

$$R_0^{(1)} = \frac{\beta}{\gamma}$$

- ▶ Universal vaccination ($x_V = 1, S = 0, V = 1$):

$$R_0^{(2)} = \frac{\beta(1 - \epsilon_V)}{\gamma}$$

Key insight: $R_0 < 1 \rightarrow$ infection dies out; $R_0 > 1 \rightarrow$ endemic persistence. Vaccination reduces R_0 by factor $(1 - \epsilon_V)$.

Effect of Present Bias (k) on System Dynamics

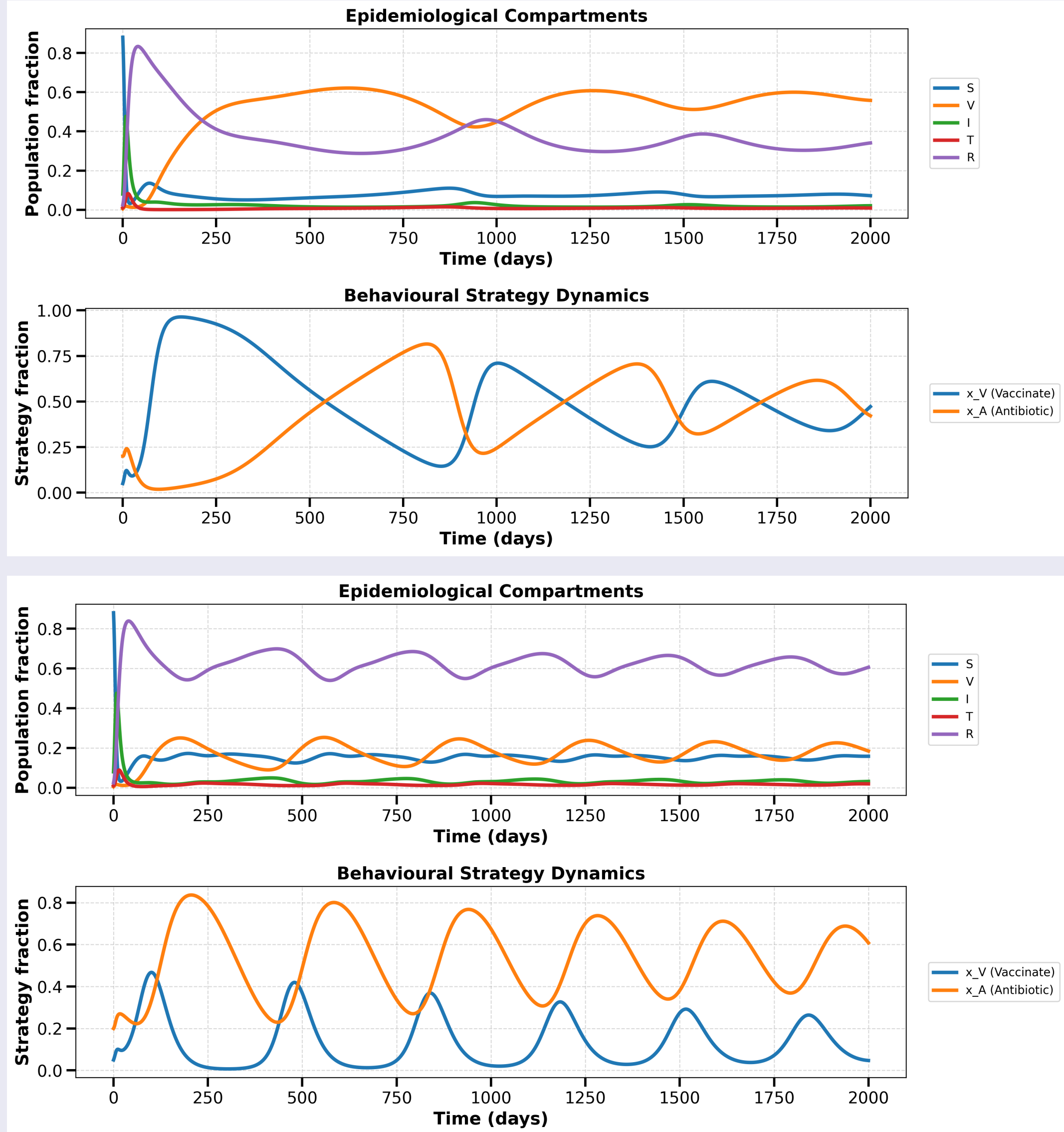


Figure 1: $k = 0.5$ (weak bias) → stable endemic equilibrium

Figure 2: $k = 1.5$ (strong bias) → infection extinction

Key insight: Moderate present bias leads to stable endemic equilibrium, while strong present bias leads to infection extinction.

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